# May the F1 Be With You: Classification of Viral Meme Success

Final Project for Applied Machine Learning (Spring 2025)

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**Abstract**

In this project, we apply machine learning to Reddit memes by modeling them as a multimodal prediction task, where image features are extracted using pretrained vision encoders and text features are extracted using transformer-based language models. We then combine these embeddings to predict meme virality, as measured by the number of upvotes. To evaluate whether multimodality improves model performance, we compare strong baseline models with deep multimodal architectures. Focusing on Reddit enables us to move away from the more heavily focused platforms and generate valuable insights. We also cluster meme embeddings across different subreddits to explore whether and how certain communities on Reddit (e.g., programmers versus engineers) differ in their sense of humor.

## 1. Introduction

Memes originated as a medium to trigger laughter, sometimes as simple as images and text or a video and a perfectly timed sound effect. These multimodal phenomena have shown promise as marketing opportunities, with viral memes encouraging customer engagement up to thirty times more than traditional branded content (Malodia et al., 2022; Paquette, 2019). Moreover, 27.5% of social media users aged 16 and older followed a meme account in 2025, reflecting its influence on adults (Kemp, 2025). Despite being acknowledged for its marketing power, the specific factors behind what makes a meme funny remain unclear, as the time of day, target audience, and number of elements in the image are only a few of the wide range of possible factors. These factors may even interact, making it even more challenging to answer the question posed by many researchers: how to predict meme virality? Its findings would not only prove valuable for marketing teams but also help these online platforms mitigate the spread of harmful content, making them safer for their users (Sah, 2025).

## 2. Background

### 2.1 General Meme Findings

With the growth of AI and the technology sector, research turns to more novel forms of marketing, one of which is meme marketing (Rolando, 2025). There is a wide range of studies on memes, from evaluating metrics of virality to exploring the ethical concerns behind sharing them. Instagram, Facebook, and Twitter (now X) are the platforms most commonly used to create these insights. Across all of this work, one thing seems clear: memes are a great way to connect emotionally with an audience and show what a brand stands for. They're also participatory, meaning the audience actually becomes part of making and sharing the memes. The kind of memes you see depends on the platform too. For example, TikTok lets users "stitch" five seconds of someone else's video into their own, which creates lots of new meme possibilities. Overall, memes let people communicate with each other—and give brands a more fun, engaging way to reach their audience.

### 2.2 Memes on Reddit

Researchers have started looking to Reddit to get new insights into what makes memes go viral (Sah, 2025). The main focus is on how users join communities for their specific interests, called subreddits. But with so many different interests, predicting virality gets tricky, since audiences—and the memes they interact with—vary a lot across the more than 2.2 million subreddits (Curry, 2026).

### 2.3 Hypothesis

We hypothesize that subreddit identity is a major driver of meme virality on Reddit, while upload time provides a weaker but potentially useful contextual signal. Additionally, we expect that combining subreddit identity, upload time, and multimodal meme content (image and text embeddings) will predict top-25% upvote performance better than metadata alone, because community context and content quality are jointly necessary for virality.

***Assumptions***

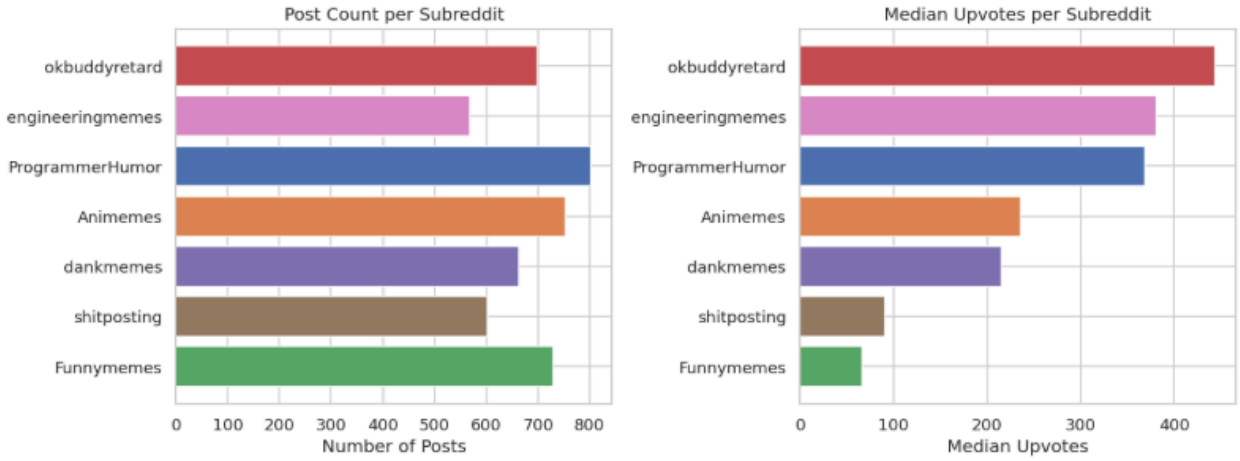
**Subreddit as a stable categorical feature.** We assume that subreddits are a stable/fixed category that will not be changed. Moreover, community norms and engagement behaviors remain relatively

consistent over the collection period. Across subreddits, median upvotes are highest in r/okbuddyretard, with a relatively similar number of posts across all seven subreddits, as shown in Figure 2.

**Upload time as a temporal feature.** We assume that upload time is a context-dependent temporal signal that indicates fluctuations in traffic. While it has little impact when looking broadly, its effect becomes stronger when time frames are introduced (e.g., morning, afternoon, and night). Therefore, we treat it as a secondary feature that has some impact on predicting vitality, but is not the main driver.

**Figure 2**

*Comparison of total posts and median upvotes across subreddits*




## 3. Methods

### 3.1 Dataset

A dataset of 4,821 Reddit memes from seven subreddits, including r/Animemes, r/dankmemes, and r/ProgrammerHumor, is used in our project. These memes were created from April 21, 2024, to April 4, 2025. Twenty-one features are included for each meme, such as the author, upvotes, and text extracted from the image using Tesseract OCR. The dataset contains no missing values across all 21 columns. An example of a meme from this dataset is displayed in Figure 1.

**Figure 1**

*Meme from the r/funnymemes subreddit with 78 upvotes*

**3.2 Dataset Analysis**

To better understand the data, we explore its textual, image, and upload time characteristics to gain insight into Reddit memes. First, each meme has metadata including upvote ratio, number of comments, and the time of day and day of the week. The latter two capture temporal engagement patterns such as morning posts attracting the highest median upvotes, while weekend posts perform slightly better overall. Second, each meme has text features, including title word count and OCR-extracted word count from the meme image. Examining the title length, having a longer or shorter title likely has little impact on virality as it has a weak negative correlation with upvotes ($r = -0.120$). This also extends to OCR length, which is negligible ($r = +0.019$). Third, each meme has sentiment features that are derived using both TextBlob and a RoBERTa-based model fine-tuned on social media text, capturing title polarity and subjectivity. However, sentiment shows near-zero correlation with upvotes across all subreddits, suggesting it has little predictive power on its own. Fourth, each meme has author features, including total karma and account age in days, both of which also show negligible correlation with upvotes ($r \approx -0.01$ to $-0.03$). Lastly, subreddit identity is treated as a categorical feature due to substantial variation in viral rates across communities, ranging from 14.1% in r/Funnymemes to 32.3% in r/ProgrammerHumor.

### 3.3 Defining a viral meme

Reddit has multiple useful metrics to determine whether a meme is viral, such as the number of comments or the ratio of upvotes to total votes. After analyzing the data, we define virality as a binary label derived by thresholding upvotes at the 75th percentile (i.e., 1,202 upvotes). To start, we find that raw upvotes are heavily right-skewed (mean = 1,504; median = 240; max = 62,256), justifying the use of a binary classification framing over direct regression. Furthermore, a majority-class classifier achieves 75% accuracy with a dummy F1 score of 0.375, serving as the performance baseline for our project.

### 3.4 Data Cleaning and Processing

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### 3.5 Modeling Approach

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## 4. Results

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## 5. Discussion

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## References


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